

Hibernate Framework MCQs

1. What is Hibernate primarily used for?

- A) Web Development
- B) Object-Relational Mapping (ORM)
- C) Mobile Application Development
- D) Game Development

Answer: B) Object-Relational Mapping (ORM)

2. Which of the following is a key feature of Hibernate?

- A) Automatic Schema Generation
- B) Direct SQL Query Execution
- C) Manual Transaction Management
- D) None of the above

Answer: A) Automatic Schema Generation

3. What does HQL stand for in Hibernate?

- A) Hibernate Query Language
- B) High Query Language
- C) Hibernate Quick Language
- D) None of the above

Answer: A) Hibernate Query Language

4. Which annotation is used to define an entity class in Hibernate?

- A) @Table
- B) @Entity
- C) @Column
- D) @Id

Answer: B) @Entity

5. What is the purpose of the SessionFactory in Hibernate?

- A) To manage database connections
- B) To create Session instances
- C) To execute HQL queries
- D) To configure Hibernate settings

Answer: B) To create Session instances

6. Which of the following is NOT a state of an entity in Hibernate?

- A) Transient
- B) Persistent
- C) Detached
- D) Active

Answer: D) Active

7. What does the @Id annotation signify in an entity class?

- A) It marks the primary key of the entity.
- B) It indicates a foreign key relationship.
- C) It specifies a unique constraint.

- D) It defines a column in the database.

Answer: A) It marks the primary key of the entity.

8. Which method is used to begin a transaction in Hibernate?

- A) session.startTransaction()
- B) session.beginTransaction()
- C) transaction.begin()
- D) session.openTransaction()

Answer: B) session.beginTransaction()

9. What is the purpose of the @GeneratedValue annotation?

- A) To specify the data type of a column
- B) To automatically generate values for the primary key
- C) To define a unique constraint
- D) To map a column to a Java field

Answer: B) To automatically generate values for the primary key

10. Which of the following is a valid way to query data using HQL?

- A) SELECT * FROM Employee
- B) FROM Employee
- C) GET Employee
- D) QUERY Employee

Answer: B) FROM Employee

11. What does the cascade attribute in a relationship annotation do?

- A) It defines the type of relationship.
- B) It specifies the behavior of related entities during operations.
- C) It indicates the database table name.
- D) It sets the fetch type for the relationship.

Answer: B) It specifies the behavior of related entities during operations.

12. Which of the following is a valid collection type supported by Hibernate?

- A) ArrayList
- B) HashMap
- C) Set
- D) All of the above

Answer: D) All of the above

13. What is the role of the Transaction interface in Hibernate?

- A) To manage database connections
- B) To handle CRUD operations
- C) To manage database transactions
- D) To configure Hibernate settings

Answer: C) To manage database transactions

14. Which of the following is true about the @OneToMany annotation?

- A) It defines a one-to-one relationship.
- B) It defines a one-to-many relationship.
- C) It defines a many-to-many relationship.

- D) It is used for mapping basic types.

Answer: B) It defines a one-to-many relationship.

15. What does the @Embeddable annotation signify?

- A) It marks a class as an entity.
- B) It indicates that a class can be embedded in another entity.
- C) It defines a primary key.
- D) It specifies a foreign key relationship.

Answer: B) It indicates that a class can be embedded in another entity.

16. Which of the following is a valid way to define a named query in Hibernate?

- A) @NamedQuery(name = "Employee.findAll", query = "FROM Employee")
- B) @Query(name = "Employee.findAll", query = "FROM Employee")
- C) @NamedQuery("Employee.findAll", "FROM Employee")
- D) @Query("Employee.findAll", "FROM Employee")

Answer: A) @NamedQuery(name = "Employee.findAll", query = "FROM Employee")

17. What is the purpose of the @ElementCollection annotation?

- A) To map a collection of entities.
- B) To map a collection of basic types or embeddable objects.
- C) To define a one-to-many relationship.
- D) To specify a primary key.

Answer: B) To map a collection of basic types or embeddable objects.

18. Which of the following is NOT a benefit of using Hibernate?

- A) Simplified database interactions
- B) Automatic schema generation
- C) Direct SQL execution
- D) Object-oriented data manipulation

Answer: C) Direct SQL execution

19. What does the @JoinColumn annotation specify?

- A) The name of the table in the database.
- B) The name of the column that acts as a foreign key.
- C) The name of the primary key column.
- D) The name of the entity class.

Answer: B) The name of the column that acts as a foreign key.

20. Which of the following is a valid way to perform a bulk update using HQL?

- A) UPDATE Employee SET salary = salary * 1.1
- B) UPDATE Employee e SET e.salary = e.salary * 1.1
- C) UPDATE Employee e WHERE e.salary > 50000
- D) All of the above

Answer: D) All of the above

